Caio César Graciani Rodrigues, PhD
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Naples, Italy

1 Education

2013 – 2017
Doctor of Science, D.Sc., in Computational Modeling
National Laboratory for Scientific Computing (LNCC), Petrópolis, RJ, Brazil
Thesis: Control and Filtering for Continuous-time Markov Jump Linear Systems with Partial Mode Information.
Advisors: Prof. Marcos Garcia Todorov and Prof. Marcelo Dutra Fragoso.

2011 – 2013
Master of Science, M.Sc., in Computational Modeling in Science and Technology
Fluminense Federal University (UFF), Volta Redonda, RJ, Brazil
Dissertation: Age Effects in the Spreading of Tuberculosis
Advisors: Prof. Thadeu Josino Penna and Prof. Aquino Lauri de Espíndola.

2007 – 2010
Major in Mathematics
Fluminense Federal University (UFF), Volta Redonda, RJ, Brazil

2 Current Position

C.C. Graciani Rodrigues is currently holding a two-year PNRR postdoctoral fellowship at the Department of Modeling and Engineering of Risk and Complexity (MERC), Scuola Superiore Meridionale (SSM), Naples, Italy.

3 Professional experience

3.1 Postdoctoral Position

- Three-year postdoctoral position at the Department of Modeling and Engineering of Risk and Complexity (MERC), Scuola Superiore Meridionale (SSM), Naples, Italy, from June 2022 to April 2025.
- From June 2017 to May 2022, C. C. Graciani Rodrigues held a level D-B research grant at LNCC, funded by the National Council for Scientific and Technological Development (CNPq/Brazil).

3.2 Invited Session (ECC 2026)

Invited Session at the European Control Conference (ECC 2026), held in Reykjavík, Iceland. The session, entitled "Robust Learning, Control, and Games under Uncertainty: Theory, Algorithms and Applications," was jointly organized by Dr. Caio César Graciani Rodrigues and Prof. Giovanni Russo (University of Salerno).

3.3 Teaching

Course on Probability and Elements of Stochastic Modeling

- *Scuola Superiore Meridionale* – Department MERC; Naples, Italy
- Jan-Feb, 2023 (12H); Jan-Feb, 2024 (12H); Jan-Feb, 2025 (12H); & Nov, 2025 (12H).
- *Brief description:* Course on probability and stochastic modeling, focusing on applications in engineering and science. Gained proficiency in probability theory, random variable models, Markov chains, and Poisson processes. Collaborated in teams on weekly assignments involving MATLAB, Phyton or R, applying theoretical concepts to real-world scenarios and developing practical problem-solving skills.

Course on Mathematics and Data Modeling

- *Scuola Superiore Meridionale* – Department MERC; Naples, Italy
- Nov-Dec, 2022 (24H); & Nov 2024 -Jan 2025 (48H)
- *Brief description:* Interdisciplinary course in mathematical modeling, data analysis, and problem-solving for real-world applications in science and engineering. Developed skills in stochastic modeling, uncertainty quantification, and risk assessment through case studies.

Undergraduate Courses: Calculus I; Linear Algebra

- *Centro Universitário de Volta Redonda* – UniFOA; Volta Redonda, RJ, Brazil
- (2012 – 2013, 2 semester per year)

Undergraduate Courses: Discrete Mathematics; Calculus I and II; Ordinary Differential Equations

- *CEDERJ Volta Redonda*; Volta Redonda, RJ, Brazil
- (2011 – 2013, 2 semester per year)

Undergraduate Courses: Analytical Geometry; Ordinary Differential Equations

- Internship at the Fluminense Federal University; Volta Redonda, RJ, Brazil
- (2011, 1 semester)

4 Skills

Languages: Portuguese – Mother tongue;
English – reading, writing, and oral (advanced);
Italian – C1
Programming Languages: Matlab, Phyton, C. Typesetting system: LaTeX.

5 Publications

5.1 Journal Articles

T. De Angelis, C. C. Graciani Rodrigues and P. Tankov. *A Model of Strategic Sustainable Investment.* http://arxiv.org/abs/2412.00986. Mathematical Finance. Available online on 20 May 2026, https://doi.org/10.1111/mafi.70033.

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *Fast detector-based H_2-control for Markov jump linear systems with multiplicative noises.* SIAM Journal on Control and Optimization, 59 (6): 4243–4267. Available online: 04 November 2021. DOI: https://doi.org/10.1137/20M1335303

M. G. Todorov, F. O. dos Santos, C. C. Graciani Rodrigues. *Homogenized First-Moment Analysis of Two-Time-Scale Positive Markov Jump Linear Systems.* Journal of the Franklin Institute. Available online: 22 January 2021. DOI: https://doi.org/10.1016/j.jfranklin.2021.01.016

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *Detector based approach for H_∞-filtering of Markov jump linear systems with partial mode information.* IET Control Theory & Applications Special Issue: Recent Advances in Control and Verification for Hybrid Systems, 13(9):1298–1308, 2019. http://ietdl.org/t/xZX83b

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *H_∞ control of continuous-time Markov jump linear systems with detector-based information.* International Journal of Control, 90(10), pages 2178–2196, 2017. https://doi.org/10.1080/00207179.2016.1238511

C. C. Graciani Rodrigues, A. L. Espíndola, and T. J. Penna. *An agent-based computational model for tuberculosis spreading on age-structured populations.* Physica. A, 428, pages 52–59, 2015. https://doi.org/10.1016/j.physa.2015.02.027

5.2 Conference Proceedings

A. Shafiei, C. C. Graciani Rodrigues and G. Russo. *Free-Energy Minimizing Policies under Generative Model Ambiguity.* https://arxiv.org/abs/2604.08222 2026 European Control Conference (ECC), pages 2006-2013

A. Del Vecchio, A. Della Pia, and C. C. Graciani Rodrigues. (2026). *A Hybrid Theoretical and Data-Driven Approach to Crowd Dynamics Modelling.* Multi-Agent Systems: 22nd European Conference, European Conference on Multi-Agent Systems (EUMAS) 2025, Bucharest, Romania, September 3–5, 2025, Proceedings, Part I (Lecture Notes in Computer Science, Vol. 16258; Lecture Notes in Artificial Intelligence). Springer Cham.

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *Mean square stability and H_2-control for Markov Jump linear systems with multiplicative noises and partial mode information.* In 57*th* IEEE Conference on Decision & Control, pages 5604–5609, Miami Beach, FL, USA,

2018. DOI: 10.1109/CDC.2018.8619184

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *H_∞ filtering for Markovian jump linear systems with mode partial information*. In 55*th* IEEE Conference on Decision & Control, pages 640–645, Las Vegas, USA, December 2016. DOI: 10.1109/CDC.2016.7798341

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *H_∞ control for continuous-time Markov jump linear systems with partial mode information*. In *XXI Congresso Brasileiro de Automática*, pages 1572–1577, Vitória, ES, Brazil, October 2016.

C. C. Graciani Rodrigues, M. G. Todorov, and M. D. Fragoso. *A bounded real lemma for continuous-time linear systems with partial information on the Markovian jumping parameters*. In 54*th* IEEE Conference on Decision & Control, pages 4226–4231, Osaka, Japan, 2015. DOI: 10.1109/CDC.2015.7402878